Numeric Results

A Novel Hybrid Fuzzy Membership Function-Based Feature Transformation for Cardiovascular Disease Prediction

Results-only documentation · Fuzzy-GBDT & Bagging Fuzzy-GBDT · tables follow the paper order

1. Setup (the way results were produced)

- **Datasets.** D1 = UCI Heart Disease (compact clinical set). D2 = Framingham Heart Study (>5000 records, 16 features = 15 input + 1 target `TenYearCHD`).
- **Split.** 70% train / 30% test on both datasets.
- **Validation.** 10-fold cross-validation + nested CV (outer 10-fold for evaluation only; inner loop for Grid Search, 144 hyperparameter combinations per outer fold).
- **Models scored.** Logistic Regression (LR), Naïve Bayes (NB), SVM, Decision Tree (DT), Neural Network (NN), GBDT, **Proposed Fuzzy-GBDT**, **Proposed Bagging Fuzzy-GBDT**.
- **Fuzzified attributes.** D1: 4 attributes; D2: 5 attributes. Hybrid MF = Gaussian + Sigmoid, $k = 3$ membership degrees per feature.
- **Stat tests.** Raw features: Mann–Whitney U. Transformed fuzzy features: Chi-square test of independence. Significance at $p < 0.05$; values shown as 0.000000 reported as $p < 0.000001$.

2. Headline Performance (test set)

Dataset	Accuracy	Precision	Recall	F1	AU-ROC
D1 (UCI)	**98%**	98%	98%	**99%**	**0.99**
D2 (Framingham)	**84.23%**	88.86%	70.49%	78.62%	**0.91**

- Best model behind these numbers: **Bagging Fuzzy-GBDT**.
- D1 > D2 on every metric → interpret as dataset-dependent (D2 broader, more heterogeneous), not uniform superiority.

3. Impact of Fuzzification — Feature-level AUC (Raw → Transformed)

Table V — Dataset One

Feature	**Raw AUC** (M-W p)	**Transformed AUC** (χ^2 p)
Age	0.3613 ($p < 0.005775$)	**0.5000** ($p = 0.000012$)
Cholesterol	0.4232 ($p = 0.000363$)	**0.5657** ($p = 0.000021$)
Resting BP	0.4338 ($p = 0.049434$)	**0.5828** ($p = 0.000233$)
Maximum Heart Rate	0.7482 ($p < 0.000001$)	**0.7933** ($p < 0.000001$)

Table VI — Dataset Two

Feature	**Raw AUC** (M-W p)	**Transformed AUC** (χ^2 p)
Age	0.5324 ($p < 0.000011$)	**0.6724** ($p < 0.000003$)
Total Cholesterol	0.5586 ($p < 0.002004$)	**0.5637** ($p < 0.000102$)
Diastolic BP	0.6294 ($p < 0.003001$)	**0.6428** ($p < 0.000648$)
Systolic BP	0.6600 ($p < 0.000001$)	**0.6800** ($p < 0.000001$)

- Every fuzzified feature's AUC increased. Largest D1 lift: Age (+0.139) & Resting BP (+0.149). Largest D2 lift: Age (+0.140).

4. State-of-the-Art Comparison

Table VII — Dataset One (% with 95% CI)

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	82 (70.89–87.11)	79 (66.54–88.00)	85 (73.33–92.90)	82 (73.47–89.66)
Naïve Bayes	81 (70.89–87.11)	78 (65.63–87.05)	87 (75.76–94.27)	82 (73.97–89.91)
SVM	93 (84.96–96.22)	90 (79.39–95.82)	96 (86.29–98.87)	93 (87.36–97.73)
Decision Tree	82 (73.33–88.88)	79 (66.71–87.53)	93 (80.81–96.78)	85 (76.77–91.59)
Neural Network	94 (86.35–96.94)	94 (85.75–98.83)	91 (80.81–96.78)	95 (88.10–98.08)
GBDT	84 (75.82–90.61)	80 (67.73–87.96)	97 (86.29–98.87)	88 (79.59–93.10)
Fuzzy-GBDT	**97** (92.34–99.40)	**99** (89.31–99.64)	**97** (89.31–99.64)	**98** (94.62–100)
Bagging Fuzzy-GBDT	**98** (92.34–99.40)	**98** (89.31–99.64)	**98** (89.31–99.64)	**99** (94.62–100)

Table VIII — Dataset Two (% with 95% CI)

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	69.48 (66.97–71.83)	60.80 (57.07–64.41)	72.61 (68.80–76.13)	66.18 (63.07–69.16)
Naïve Bayes	65.04 (62.46–67.49)	64.41 (58.79–69.65)	33.57 (29.80–37.56)	44.13 (39.95–48.17)
SVM	74.49 (72.10–76.71)	65.42 (61.82–68.86)	80.57 (77.10–83.61)	72.21 (69.34–74.94)
Decision Tree	77.40 (75.10–79.52)	71.00 (67.27–74.47)	76.15 (72.47–79.48)	73.49 (70.58–76.25)
Neural Network	73.69 (71.28–75.93)	67.11 (63.24–70.77)	70.67 (66.79–74.27)	68.85 (65.71–71.83)
GBDT	81.01 (78.86–83.00)	**89.61** (86.43–92.27)	66.20 (62.26–70.03)	76.14 (72.75–78.72)
Fuzzy-GBDT	83.72 (81.66–85.57)	**90.33** (87.14–92.79)	67.67 (63.71–71.39)	77.37 (74.40–80.20)
Bagging Fuzzy-GBDT	**84.23** (82.20–86.05)	88.86 (85.62–91.45)	**70.49** (66.61–74.10)	**78.62** (75.75–81.31)

- D1: proposed models top accuracy/precision/F1. D2: Bagging Fuzzy-GBDT best on accuracy, recall, F1; Fuzzy-GBDT best on precision.

5. AUROC of Proposed Models

Model	D1 AUC	D2 AUC
Fuzzy-GBDT	0.98	0.907
Bagging Fuzzy-GBDT	**0.99**	**0.911**

6. MCC and Cohen's Kappa

Dataset	Model	MCC	Kappa
D1	Logistic Regression	0.6017	0.5993
	Naïve Bayes	0.6034	0.5979
	SVM	0.8463	0.8444
	Decision Tree	0.6519	0.6414
	Neural Network	0.8686	0.8678
	GBDT	0.7031	0.6851
	Fuzzy-GBDT	**0.9558**	**0.9558**
	Bagging Fuzzy-GBDT	**0.9558**	**0.9558**
D2	Logistic Regression	0.3924	0.3873
	Naïve Bayes	0.2468	0.2219
	SVM	0.4998	0.4907
	Decision Tree	0.5391	0.5381
	Neural Network	0.4612	0.4608
	GBDT	0.6456	0.6281
	Fuzzy-GBDT	0.6671	0.6505
	Bagging Fuzzy-GBDT	**0.6749**	**0.6637**

7. Overfitting Check (train vs. validation, 100% training data)

Dataset	Train Acc	Val Acc	Gap
D1	0.9930	0.9643	0.0287
D2	0.8809	0.8401	0.0408

- Small gaps on both datasets → no severe overfitting.

8. Membership-Function Ablation — Accuracy (%)

Table IX — Triangular vs. Trapezoidal vs. Proposed Hybrid

Dataset	Model	Triangular	Trapezoidal	Hybrid (Proposed)
D1	Fuzzy-GBDT	93.50	71.05	**97.00**
D1	Bagging Fuzzy-GBDT	92.85	68.74	**98.00**
D2	Fuzzy-GBDT	82.30	67.89	**83.72**
D2	Bagging Fuzzy-GBDT	83.04	63.08	**84.23**

- Hybrid MF wins in every row; trapezoidal is consistently weakest.

9. 10-Fold CV Summary (Bagging Fuzzy-GBDT)

Dataset	Acc	Prec	Recall	F1	AUC
D1	98%	98%	97%	98%	0.99
D2	≈84%	high/stable	stable	stable	—

10. Nested CV — Fold-wise (Bagging Fuzzy-GBDT)

Dataset One

Fold	Acc	Prec	Recall	F1
1	98.10	98.90	97.20	98.00
2	94.80	94.10	95.20	94.60
3	98.20	99.10	97.30	98.10
4	99.00	99.20	98.70	98.95
5	97.20	97.10	96.80	97.05
6	94.40	96.70	91.60	94.10
7	98.10	97.20	98.90	98.00
8	96.40	95.50	97.10	96.30
9	98.20	97.30	98.80	98.10
10	93.40	96.90	89.90	93.20
Mean±SD	**96.78±1.94**	**97.20±1.61**	**96.15±3.08**	**96.64±2.00**
95% CI	96.02–98.30	96.38–98.80	94.28–98.52	95.47–98.49

Dataset Two

Fold	Acc	Prec	Recall	F1
1	84.00	89.00	67.00	76.50
2	82.50	87.80	63.80	73.90
3	83.20	88.30	65.00	74.90
4	84.00	89.70	66.20	76.20
5	83.70	90.80	63.50	74.70
6	82.90	89.90	58.50	71.00
7	84.50	88.50	69.00	77.70
8	85.00	88.90	71.00	79.00
9	85.50	89.80	74.20	81.20
10	83.40	88.40	63.00	73.60
Mean±SD	**83.87±0.94**	**89.11±0.92**	**66.12±4.47**	**75.87±2.93**
95% CI	83.20–84.54	88.45–89.77	62.92–69.32	73.77–77.97

- D1 folds tight (low SD) → stable. D2 recall most variable (SD 4.47) → harder positive-case detection.

All values transcribed from the source manuscript. D1 = UCI Heart Disease; D2 = Framingham Heart Study. Bold = proposed models / best per block. Results subject to further external and clinical validation.